

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester VI Examination, 2013-14

Paper: GLB-604 — Economic Geology

Time: 3 Hours

Full Marks: 70

Ref. No.: BS/Sem VI/228

*Note: The figures in the right-hand margin indicate marks. Answer **five** questions in all, selecting **two** each from Sections A and B. Section-C is compulsory.*

SECTION — A

Question 1. Discuss the origin of petroleum and also write its important distribution in India. [14]

Question 2. Write an essay on hydrothermal process of formation of ore deposit with Indian example. [14]

Question 3. What is placer deposit? Discuss the mechanical concentration process of formation of ore deposit with Indian example. [14]

Question 4. Enlist the important minerals/rocks raw material used in refractory industry. Write their chemical composition, specification (if any) and important distribution in India. [14]

SECTION — B

Question 5. Give the chemical compositions, physical properties, specification for different industrial uses and important distribution in India of **any two** of the following: [$2 \times 7 = 14$]

(a) Halite

(b) Phosphorite

(c) Pyrite

Question 6. Write note on **any two** of the following: [$2 \times 7 = 14$]

(a) Very common forms and structures of ore deposits

(b) Sedimentation process

(c) Contact metasomatic process

Question 7. Answer the following: [$2 \times 7 = 14$]

(a) Discuss the important distribution of coal (bituminous and lignite) in India.

- (b) Write a note on building stone.

SECTION — C

Question 8. Answer the following: [14]

- (a) Give neat sketch of supergene sulphide enrichment process. [4]
- (b) Differentiate between contact metamorphism and contact metasomatism. [2]
- (c) Name the host rock associated with—
- (i) Amorphous variety of magnesite
 - (ii) Chromite deposit
 - (iii) Diamond
 - (iv) Mica deposit

[4 × $\frac{1}{2}$ = 2]

- (d) Write in brief the different stages of coal formation. [3]
- (e) Write in brief the basic relation between paragenesis and zoning. [3]